The Project-Capacity Selection Problem in Place-Based Funding

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Abstract

This article promotes a seeded research idea into an auditable manuscript. It treats Regional science as a design problem: first identify the literature gap, then choose the strongest data and regression design the evidence can support, and only then publish a final PDF. The immediate contribution is a reproducible paper scaffold with claims, sources, data manifests, and reviewer automation. The substantive contribution is a design ladder for the project-capacity selection problem in place-based funding: source mapping, candidate data acquisition, staged descriptive results, and escalation to hierarchical, spatial, event-study, synthetic-difference, causal-forest, or double-machine-learning estimators only when the data justify them.

1 Introduction

The farm seed behind this paper is not treated as a finished insight; it is treated as a wager about where the literature may still have an under-measured empirical object. Recent regional and applied social-science work shows why local capacity, institutions, and uneven development matter (Diemer et al., 2022; Iammarino et al., 2019; Rodriguez-Pose, 2018). The missing step in many automated paper systems is the conversion from an idea to an inspectable research object. This manuscript therefore makes the conversion visible: topic, literature map, data plan, empirical ladder, claims ledger, uncertainty register, and final PDF gate.

The contribution is intentionally narrow. It does not claim that the seed already estimates a causal effect. It claims that Regional science papers can be moved through a disciplined pipeline where bibliography, claims, data artifacts, and reviewer questions are produced together (OpenAlex, 2026; Peng, 2011). That matters because the most attractive model is often not the right model; the right model is the strongest one the data can defend.

2 Literature Gap

The first gap is infrastructural. Open-access discovery tools now make it easier to find candidate literatures, but the link between literature search and empirical design remains fragile when the paper is generated quickly (OpenAlex, 2026; Kitchin, 2014). The second gap is methodological. Regional and policy literatures often need to decide whether the object is a descriptive conversion process, a treatment effect, a spatial spillover, or a heterogeneous-response problem (Barca, 2009; Diemer et al., 2022).

This article fills that gap by making the design choice explicit. It asks what a reviewer would need to see before accepting stronger claims: a source ledger, a data manifest, transparent uncertainty, and section-level references throughout the manuscript. The claim is modest but useful: a

*Draft date: May 14, 2026. Built by the scriptor farm promotion pipeline after reviewer automation.

paper farm should not merely farm titles; it should farm auditable, citation-bearing, reviewer-facing research packages (Iammarino et al., 2019; Rodriguez-Pose, 2018).

3 Data

The proof-of-concept data layer is deliberately clean. It keeps only derived artifacts: a topic profile, a stage-count table, an OpenAlex search manifest, and a literature map. The declared data types for this job are administrative place panels, programme awards, procurement notices, contracts, spatial boundaries, local need and capacity indicators. Raw downloads are excluded from the final output and should be deleted after derived manifests and reproducible instructions have been written.

The OpenAlex layer is represented as a transparent acquisition manifest rather than as hidden context (OpenAlex, 2026). For a production run, each search query would be executed against the works endpoint, open-access full-text URLs would be normalized, and each retained article would receive a source record. Until that acquisition succeeds, the results below remain pipeline-readiness diagnostics rather than substantive estimates (Kitchin, 2014; Peng, 2011).

4 Empirical Strategy

The empirical strategy is a ladder rather than a single preferred regression. It begins with stage conversion rates and missingness diagnostics. It moves to hierarchical partial pooling only when observations are nested in places, institutions, papers, labs, courts, or schools with enough repeated outcomes (Gelman and Hill, 2007). It uses spatial models when adjacency, distance, or network exposure is part of the estimand (Anselin, 1988).

For causal questions, the ladder escalates only when the design warrants it. Event study or synthetic difference estimators are appropriate when treatment timing is plausibly as-good-as-random conditional on pre-trends (Arkhangelsky et al., 2021). Double machine learning and causal forest designs are appropriate when nuisance functions and heterogeneity are central and sample size is adequate (Chernozhukov et al., 2018; Wager and Athey, 2018). Where data do not identify assignment or delivery, the paper should use partial identification and sensitivity analysis instead of point-identifying claims it cannot defend (Rosenbaum, 2002; Imbens and Rubin, 2015).

5 Results

The current run produces a reviewer-gated article package rather than a substantive treatment estimate. Table 1 records the objects produced from the seed and the next empirical escalation. This is a result about pipeline state: the job is no longer merely an idea once it has a manuscript, citations, claims ledger, source ledger, data manifest, and reviewer-question pack (Peng, 2011).

Table 1: Promotion-stage artifacts

Stage	Count	Interpretation
Seeded topic	1	Farm idea selected for promotion
Declared data types	6	Candidate acquisition surfaces
Search queries	3	Literature-discovery starting points
Method families	5	Regression designs requiring gates
Reviewer gates	9	Claims, data, methods, citations, and audit checks

The design table in Table 2 keeps the regression choices honest. It records the candidate method families for this topic and the gate that must be passed before each family is used in a future empirical version. This avoids the common failure mode where a sophisticated estimator appears before the data structure has earned it (Chernozhukov et al., 2018; Wager and Athey, 2018; Arkhangelsky et al., 2021).

Table 2: Method gates for later empirical versions

Candidate method	Gate before use
Spatial Models	Use only when data structure and identification support it
Stage Conversion Models	Use only when data structure and identification support it
Event Studies	Use only when data structure and identification support it
Synthetic Controls	Use only when data structure and identification support it
Causal Forests When Covariates Justify Them	Use only when data structure and identification support it

6 Reviewer Automation and Limits

The reviewer layer checks that citations appear in the text, bibliography keys are used, claims have sources, weak claims do not carry causal language, data artifacts are declared, and reviewer questions can be generated from the ledger. These checks do not make the argument true, but they make the argument easier to audit (Peng, 2011; OpenAlex, 2026).

The main limitation is that this promoted draft should not be mistaken for a completed empirical article until the subject-specific data pull has been executed and inspected. The strongest version of the paper will replace the readiness diagnostics with real estimates, figures, and robustness checks. The present version is still useful because it creates the article-shaped container that lets those estimates arrive without breaking the claims, data, and reference rules (Kitchin, 2014; Rosenbaum, 2002).

7 Conclusion

This paper turns a farm seed into a reviewer-facing research package for Regional science. Its contribution is a modular path from idea generation to auditable paper output: sources, claims, data, methods, LaTeX, PDF, and public shelf. The next iteration should execute the OpenAlex and subject-data acquisition, then rerun the same reviewer automation after replacing readiness diagnostics with validated empirical results (Diemer et al., 2022; Iammarino et al., 2019; Peng, 2011).

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